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Exact Computation for the Normalized Power Prior

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Statement of Authorship

I hereby declare that the thesis submitted is my own work. All direct or indirect sources used are acknowledged as references. I further declare that I have not submitted this thesis at any other institution in order to obtain a degree.

Acknowledgements

I would like to thank [many people]

Abstract

In statistical modelling with historical data, a central challenge is how to effectively integrate prior evidence with current observations. This issue is particularly pressing in clinical trials and medical applications, where using historical data can improve both the efficiency and robustness of inference. A natural Bayesian strategy for this purpose is the use of informative priors. Among them, the normalised power prior has attracted significant interest, as it provides a principled way to control the influence of historical information. The method introduces a power parameter, $\eta \in [0, 1]$, that governs the degree of borrowing: $\eta = 0$ discards the past data while $\eta = 1$ fully incorporates it. This flexibility makes the framework highly appealing in practice. Despite its advantages, the power prior posterior distribution is doubly intractable, as it requires evaluation of a normalising constant that depends on η. Existing approaches rely on approximate inference, which often lack theoretical guarantees and explicit convergence properties. To overcome these limitations, we propose an exact MCMC algorithm based on pseudo-marginal methods. The key idea is to replace the intractable ratio of normalising constants with an unbiased estimator. After finding useful identities for the derivative of the log normalising constant, we construct unbiased estimators of log-ratios through Monte Carlo integration, enabling exact acceptance probabilities, by employing the Poisson Estimator Method, within a Metropolis-Hastings or Barker's scheme. We evaluate the method on Gaussian models under varying degrees of compatibility between historical and current data. Results demonstrate that the algorithm accurately reflects data conflict by shrinking η when the datasets are incompatible, while recovering correct posterior behaviour when they are aligned. Comparisons with analytical baselines confirm that our method achieves reliable estimates of both model parameters and the power parameter, with efficiency suitable for practical use. This provides a rigorous, exact alternative to approximate approaches, offering stronger guarantees for applications such as adaptive clinical trial design.

Keywords: Bayesian Inference; Power Prior; Doubly Intractable Distributions; Exact MCMC; Historical Data Analysis

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1 Introduction

`Luiz:` Second-to-last thing to be written. Don't sweat it. Leave it to *Papa*

- Describe NPP and notation; motivation with historical data (MANY CITATIONS).
- The NPP as a doubly-intractable problem (the prior is intractable)
- Mention that we start with approximate methods and then move on to exact stuff

In statistical modelling with historical data, a central challenge is effectively integrating information from past studies with current data. This is particularly relevant in clinical trials and medical research, where leveraging historical data can improve the efficiency and robustness of statistical inferences. A natural approach in this setting is the use of *informative priors*, which incorporate prior knowledge into the analysis, thereby enhancing parameter estimation and decision-making. `Adame:` Some references here would be nice.

Among the various methods for constructing informative priors, *power priors* (Ibrahim et al., 2015) have gained significant attention. They allow for controlled borrowing of information from historical data by adjusting a power parameter, η, which typically ranges between 0 and 1. When $\eta = 0$, no historical information is incorporated, whereas $\eta = 1$ fully integrates the historical data into the current analysis. `Adame:` I would like to introduce the NPPs issues here and then talk about the motivation for the exact inference. Didn't find a good way to do it.

Despite their advantages, the posterior distribution in these models is often *doubly intractable*, making standard Markov Chain Monte Carlo (MCMC) methods challenging to apply. Current approaches rely on *approximate methods* (Adame; Carvalho, 2025; Carvalho; Ibrahim, 2021), which lack theoretical guarantees and explicit convergence bounds. This work addresses this gap by developing an *exact MCMC algorithm* (Bhattacharya, 2021; Murray; Ghahramani; MacKay, 2012) capable of efficiently sampling from these complex posterior distributions.

2 Background

2.1 Bayesian Inference

2.1.1 Principles

2.1.2 Monte Carlo Methods

2.2 Standard MCMC Methods

`Adame:` Since I know you will ask me what do I mean by standard: we can surely think of a better name

2.2.1 Gibbs Sampling

2.2.2 Metropolis-Hastings

2.3 Doubly Intractable Distributions

2.4 The Normalized Power Prior

2.4.1 Formulation

Consider the following Bayesian schema:

$$p(\theta \mid D) = \frac{L(D \mid \theta)\pi_0(\theta)}{\int_\Theta L(D \mid t)\pi(t)\,\mathrm{d}t} = \frac{L(D \mid \theta)\ \overbrace{\pi_0(\theta)}^{\text{Initial prior}}}{m(D)}. \tag{2.1}$$

Definição 2.4.1 (Power prior)**.** *Let $D_0 = \{d_{01}, d_{02}, \ldots, d_{0N_0}\}$, $d_{0i} \subseteq \mathcal{X}^p$ be the historical data and let $L_0(D_0 \mid \theta)$ be a likelihood function assumed to be finite for all $\theta \in \Theta \subseteq \mathbb{R}^p$. Then, the power prior is defined as*

$$\pi_\eta(\theta \mid D_0) \propto L_0(D_0 \mid \theta)^\eta \pi_0(\theta), \tag{2.2}$$

where $\pi_0(\theta)$ is the initial prior distribution of θ and η is the (fixed) power parameter.

Then, when we want to accomodate the uncertainty in the choice of η, we treat it as a random variable and assign a prior distribution to it, indexed by φ. In this case, we find the *unnormalized* power prior as

$$\pi^\star(\theta, \eta \mid D_0, \varphi) = \pi_\eta(\theta \mid D_0)\pi_A(\eta \mid \varphi) \propto L_0(D_0 \mid \theta)^\eta \pi_0(\theta)\pi_A(\eta \mid \varphi). \tag{2.3}$$

In order to compute the posterior distribution of θ and η, we need to find the normalizing constant (from (2.2)) $c_0(\eta)$, which is given by

$$c_0(\eta) = \int_\Theta L_0(D_0 \mid t)^\eta \, \mathrm{d}P_0(t) \tag{2.4}$$

and the marginal posterior distribution of η as, for a new dataset D,

$$p(\eta \mid D, D_0, \varphi) \propto \frac{\pi_A(\eta \mid \varphi)}{c_0(\eta)} \int_\Theta L(D \mid t) L_0(D_0 \mid t)^\eta \, \mathrm{d}P_0(t). \tag{2.5}$$

Notice that, since both $c_0(\eta)$ and $\int_\Theta L(D \mid t)L_0(D_0 \mid t)^\eta \, \mathrm{d}P_0(t)$ are intractable, we call this a doubly-intractable posterior. Previous work (Adame; Carvalho, 2025; Carvalho; Ibrahim, 2021) have focused on emulating $\log c_0(\eta)$ and $\log c'(\eta) = \mathbb{E}_{\pi_\eta}[\log L_0(D_0 \mid \theta)]$ through non-parametric models, which has proven to be more effective than common interpolation methods—particularly when incorporating shape constraints.

Sampling from these posteriors is challenging due to the intractability of standard M-H acceptance probabilities, making approximate methods like auxiliary variable techniques computationally expensive and lacking convergence guarantees; an alternative is unbiased event constructions, which enable exact sampling without evaluating normalizing constants.

For instance, applying Barker's algorithm for MCMC (Vats et al., 2021) to sample from a target distribution $\pi(x) \propto \pi'(x)$ with proposal density $q(x \mid y)$ has become feasible through Bernoulli factory methods. These methods enable the construction of events with probability $\alpha_B(x, y)$ when

$$\alpha_B(x, y) = \frac{\pi(y)q(y \mid x)}{\pi(x)q(x \mid y) + \pi(y)q(y \mid x)} = \frac{\pi'(y)q(y \mid x)}{\pi'(x)q(x \mid y) + \pi'(y)q(y \mid x)} \tag{2.6}$$

cannot be directly evaluated (Gonçalves; Łatuszyński; Roberts, 2017).

2.4.2 Properties

2.4.3 Other Historical Data Modelling Approaches

2.5 Unbiased MCMC Methods

2.5.1 Barker's Algorithm for Exact Inference

Barker's algorithm (Barker, 1965) provides an alternative to Metropolis-Hastings that is amenable to Bernoulli factory methods. For a target distribution $\pi(\theta)$ and proposal

$q(\theta'|\theta)$, Barker's acceptance probability is:

$$\alpha_B(\theta, \theta') = \frac{\pi(\theta')q(\theta|\theta')}{\pi(\theta)q(\theta'|\theta) + \pi(\theta')q(\theta|\theta')} \tag{2.7}$$

While Barker's algorithm has lower efficiency than Metropolis-Hastings in the Peskun ordering, its acceptance probability has a crucial property: it can be expressed as a ratio of products. This enables the use of *Bernoulli factory* methods to implement the acceptance decision without explicitly computing α_B.

2.5.1.1 The Two-Coin Algorithm

The fundamental building block is the two-coin algorithm (Gonçalves; Łatuszyński; Roberts, 2017). Suppose we can factorize the acceptance odds as:

$$\frac{\pi(\theta')q(\theta|\theta')}{\pi(\theta)q(\theta'|\theta)} = \frac{c(\theta', \theta)p(\theta', \theta)}{c(\theta, \theta')p(\theta, \theta')} \tag{2.8}$$

where $c(\cdot, \cdot)$ is tractable and $p(\cdot, \cdot) \in [0, 1]$ is a probability we can simulate.

Then, Barker's acceptance probability becomes:

$$\alpha_B(\theta, \theta') = \frac{c(\theta', \theta)p(\theta', \theta)}{c(\theta', \theta)p(\theta', \theta) + c(\theta, \theta')p(\theta, \theta')} \tag{2.9}$$

The two-coin algorithm generates this probability without computing it explicitly:

Algorithm 1: Two-coin algorithm Just the idea

```
Flip coins with probabilities p(θ', θ) and p(θ, θ');
if both heads or both tails then
    return to step 1;
else if first heads and second tails then
    accept (return 1);
else if first tails and second heads then
    reject (return 0);
```

This procedure produces an acceptance event with exactly probability $\alpha_B(\theta, \theta')$.

2.5.2 Divide-and-Conquer Bernoulli Factory (DCBF)

For models with n observations, the likelihood factorizes as $\pi(\theta|D) \propto \prod_{i=1}^{n} f_i(\theta)$. The vanilla two-coin algorithm requires flipping a coin with probability $p(\theta) = \prod_{i=1}^{n} p_i(\theta)$. As n grows, $p(\theta)$ shrinks exponentially, causing the expected number of loops (ENL) to grow as $O(e^{cn})$ for some constant $c > 0$. This exponential cost is prohibitive even for moderate n

2.5.2.1 The Coin Merge Procedure

The DCBF addresses this by hierarchically merging factor coins. For two factors with odds h_1, h_2 and corresponding probabilities $r_j = h_j/(h_j + g_j)$, we can merge them into a single coin with probability $r_0 = h_0/(h_0 + g_0)$ where $h_0 = h_1 h_2$.

Algorithm 2: Coin merge algorithm Just the idea (again)

```
Flip coins with probabilities r_1 and r_2;
if both return the same value (both 0 or both 1) then
    return that value;
else
    return to step 1;
end
```

2.5.2.2 Hierarchical Factorization

Given n factors, we organize them in a balanced binary tree of depth $\ell = \lceil \log_2 n \rceil$.

At the leaves, we place individual factors (or small batches). At each internal node, we merge the coins from its two children.

Algorithm 3: Divide-and-Conquer Bernoulli Factory

```
function FlipDCBF({f_1, ..., f_n}, level k)
    if n = 1 then
        return Flip2Coin(f_1);
    end
    Split factors into {f_1, ..., f_{n/2}} and {f_{n/2+1}, ..., f_n};
    c_1 ← FlipDCBF(left half, k + 1);
    c_2 ← FlipDCBF(right half, k + 1);
    return MergeCoins(c_1, c_2);
end
```

2.5.2.3 Computational Complexity

(Stumpf-Fétizon; Gonçalves, 2025) prove that under regularity conditions, the expected cost of the DCBF scales as:

$$\mathbb{E}[\text{Cost}] = O(4^\ell \cdot \rho(n, \ell)) \tag{2.10}$$

where $\rho(n, \ell)$ is the cost of the two-coin algorithm at the leaves with batch size $n/2^\ell$.

For posterior concentration at rate $n^{-1/2}$ with optimal proposal scaling:

- If using likelihood ratio estimators: $\rho(n, \ell) = O(\sqrt{n/2^\ell})$

- Setting $4^\ell \propto n$ gives overall cost $O(n^{3/2})$

This is polynomial, compared to the exponential $O(e^{cn})$ cost of vanilla two-coin.

Moreover, this matches or improves upon pseudo-marginal algorithms, which typically require $O(n^2)$ cost.

2.5.3 Pseudo-marginal Metropolis-Hastings

2.6 Function Emulation Methods

2.6.1 Gaussian Processes

2.6.2 Generalized Additive Models

2.6.3 Shape-constrained Variants

3 Approximate inference with noisy-MCMC

Current approaches:

- Function emulation: (Adame; Carvalho, 2025; Carvalho; Ibrahim, 2021)

Luiz: (Han et al., 2023) (Carvalho; Ibrahim, 2021) and references therein

Luiz: Need to carefully read, cite and discuss (Park; Haran, 2020). Do their results hold for SCAM, for instance?

Our contributions in this area:

- Do MaL estimation and talk about GP and SCAM;
- Importance sampling to simplify and improve estimation of the NPP normalising constant (including control variates);
- Showcase on moderate size examples (including NIG linear regression as a baseline). Show that it works and then show how to break it

3.1 Emulating the normalizing constant

Adame: Here I'd like to present the formulation of the problem, what exactly I mean by emulating in this case.

3.1.1 Consequences

Adame: Here I'd like to talk about how appoximations w/o uncertainty quantification might lead to unstrustworthy results

3.1.2 Introducing shape-constraints

Adame: Show how to plug our scenario into these models

3.2 Current approaches for the NPP

Adame: Discuss (Han et al., 2023) and (Carvalho; Ibrahim, 2021) - and any other state-of-art

3.3 Doubly Intractable Settings

Adame: Discuss (Park; Haran, 2020)

Doubly intractable distributions arise in Bayesian inference when the likelihood takes the form

$$\pi(\theta \mid x) \propto p(\theta)\frac{h(x \mid \theta)}{Z(\theta)}, \tag{3.1}$$

where the normalizing constant $Z(\theta)$ depends on the parameter θ and cannot be evaluated pointwise, as discussed in detail on seciton 2.3. This setting is common in models such as exponential random graph models, Markov point processes, and, more generally, in likelihoods derived from Gibbs measures. In the context of Normalized Power Priors (NPP), this issue naturally appears due to the presence of parameter-dependent normalizing constants, making standard Bayesian computation challenging.

A number of Markov chain Monte Carlo (MCMC) methods have been proposed to address this problem, including auxiliary variable approaches such as the exchange algorithm and the double Metropolis–Hastings (DMH) algorithm. While these methods can target the exact posterior distribution, they typically require simulating auxiliary variables at each iteration, which becomes computationally prohibitive in high-dimensional or complex models.

An alternative strategy is proposed by Park and Haran (2020), who introduce a function emulation approach to approximate the intractable normalizing constant. Their method replaces repeated Monte Carlo estimation of $Z(\theta)$ with a two-stage surrogate model. First, importance sampling is used to obtain noisy estimates of $Z(\theta)$ at a finite set of parameter values. Then, a Gaussian process (GP) is fitted to the log-normalizing function, providing a fast interpolator that can be evaluated within an MCMC algorithm. This leads to two variants: one that emulates only the normalizing function (NormEm), and another that emulates the entire likelihood (LikEm).

The main advantage of this approach is computational efficiency. By shifting the costly Monte Carlo approximations to a precomputation stage, the resulting MCMC algorithm avoids repeated simulation of auxiliary variables, yielding substantial speed-ups compared to exact or pseudo-marginal methods. Moreover, the approach is naturally parallelizable and extends beyond doubly intractable settings to more general problems involving expensive likelihood evaluations.

However, this efficiency comes at the cost of asymptotic exactness. The resulting Markov chain targets an approximate posterior distribution, with an error that depends on both the number of importance samples and the number of design points used to train the Gaussian process. While theoretical bounds on the total variation distance are provided, they do not yield explicit rates, making it difficult to calibrate the approximation in practice.

Several additional limitations remain. First, the method relies on importance sampling estimates, which can exhibit high variance in moderate to high-dimensional settings, potentially degrading the quality of the surrogate. Second, the Gaussian process approximation assumes a smooth structure for the log-normalizing function and typically ignores heteroskedasticity induced by Monte Carlo noise. Third, the selection of design points (particles) is critical but performed using heuristic procedures such as short runs of DMH or approximate Bayesian computation, without a principled adaptive strategy. Finally, the approach does not scale well with the dimension of the parameter space due to the cubic complexity of Gaussian process regression and the increasing difficulty of covering the parameter space.

These limitations suggest several directions for future research. In particular, combining surrogate-based approximations with exact correction mechanisms, such as pseudo-marginal or delayed-acceptance schemes, could potentially recover asymptotic exactness while retaining computational gains. Additionally, replacing importance sampling with more robust estimators, and incorporating adaptive or local surrogate models, may improve both stability and scalability. Such developments are particularly relevant in the context of NPP models, where efficient and reliable inference remains an open challenge.

4 Exact inference for the Normalized Power Prior

In this section, we develop exact Markov Chain Monte Carlo methods for inference with the normalized power prior (NPP). The key challenge lies in handling the intractable normalizing constant, which we address through Russian-roulette estimators that preserve the exactness of our sampling scheme.

4.1 Barker's Method via Poisson Estimator

For the NPP, the joint posterior is:

$$\pi(\theta, \eta | D, D_0) \propto \frac{1}{c_0(\eta)} L(D|\theta) L_0(D_0|\theta)^\eta \pi_0(\theta) \pi_A(\eta) \tag{4.1}$$

The challenging term is the normalizing constant:

$$c_0(\eta) = \int_\Theta L_0(D_0|\theta)^\eta \pi_0(\theta) \, \mathrm{d}\theta \tag{4.2}$$

4.1.1 Updating θ given η

`Adame:` Here I suggest substituting to regular MH

For updating θ, the full conditional is:

$$\pi(\theta | \eta, D, D_0) \propto L(D|\theta) L_0(D_0|\theta)^\eta \pi_0(\theta) \tag{4.3}$$

Since $L_0(D_0|\theta) = \prod_{i=1}^{N_0} f_i(\theta|D_{0i})$, we can use DCBF. For instance, when we have Gaussian data: $f_i(\theta|y_i) = \mathcal{N}(y_i|\theta, \Sigma)$.

The ratio form for Barker's algorithm is:

$$\frac{\pi(\theta'|\eta)}{\pi(\theta|\eta)} = \frac{L(D|\theta')}{L(D|\theta)} \cdot \frac{\pi_0(\theta')}{\pi_0(\theta)} \cdot \prod_{i=1}^{N_0} \left(\frac{f_i(\theta'|D_{0i})}{f_i(\theta|D_{0i})} \right)^\eta \tag{4.4}$$

The first two ratios are tractable. For the product term, we construct factor coins using likelihood ratios, which benefit from posterior concentration.

4.1.2 Updating η given θ

Updating η is more subtle because the full conditional involves $c_0(\eta)$:

$$\pi(\eta | \theta, D, D_0) \propto \frac{\pi_A(\eta)}{c_0(\eta)} L_0(D_0|\theta)^\eta \tag{4.5}$$

The ratio becomes:

$$\frac{\pi(\eta'|\theta)}{\pi(\eta|\theta)} = \frac{\pi_A(\eta')}{\pi_A(\eta)} \cdot \frac{c_0(\eta)}{c_0(\eta')} \cdot L_0(D_0|\theta)^{\eta'-\eta} \tag{4.6}$$

The term $c_0(\eta)/c_0(\eta')$ is intractable but can be estimated using

$$\frac{\mathrm{d}}{\mathrm{d}\tau} \log c_0(\tau) = \mathbb{E}_{\theta\sim\pi_\tau(\theta)} \left[\log L_0(D_0 \mid \theta)\right], \tag{4.7}$$

where $\pi_\tau(\theta) \propto L_0(D_0 \mid \theta)^\tau \pi_0(\theta)$ is the power posterior at temperature τ.

Integrating this identity from η to η' yields the log-ratio:

$$\Delta = \log \frac{c_0(\eta')}{c_0(\eta)} = \int_\eta^{\eta'} \mathbb{E}_{\theta\sim\pi_\tau(\theta)} \left[\log L_0(D_0 \mid \theta)\right] \mathrm{d}\tau\,. \tag{4.8}$$

4.1.3 Practical implementation

In practice, we replace the integral Δ with a simple Monte Carlo estimation. This is a unbiased estimator for the actual integral:

$$\widehat{\Delta}_a(\eta,\eta') = \frac{(\eta'-\eta)}{K} \sum_{k=1}^{K} \mathbb{E}_{\theta\sim\pi_{\tau_k}(\theta)} \left[\log L_0(D_0 \mid \theta)\right], \text{ and so} \tag{4.9}$$

$$\widehat{\Delta}_b(\eta,\eta') = \frac{(\eta'-\eta)}{M\cdot K} \sum_{k=1}^{K} \sum_{m=1}^{M} \log L_0\Big(D_0 \mid \widetilde{\theta}_{m_k}\Big), \tag{4.10}$$

where $\tau_k \sim \mathrm{Uniform}(\eta,\eta')$, and $\left(\widetilde{\theta}_{m_k} \mid \tau_k\right) \sim \pi_{\tau_k}(\theta)$.

Thus, for a single sample $\tau_\star \sim \mathrm{Uniform}(\eta,\eta')$, and $\left(\widetilde{\theta}_\star \mid \tau_\star\right) \sim \pi_{\tau_\star}(\theta)$ we have:`Adame:` Because it doesn't depend on the variance

$$\widehat{\Delta}(\eta,\eta') = (\eta'-\eta) \log L_0\Big(D_0 \mid \widetilde{\theta}_\star\Big) \tag{4.11}$$

4.1.4 Producing an unbiased estimator via Poisson Estimator

As discussed in (Stumpf-Fétizon; Gonçalves, 2025), we can produce an unbiased estimator $\widehat{I}$ such that $\mathbb{E}_{K,(\tau_\star,\theta_\star)_1,\ldots,(\tau_\star,\theta_\star)_K}\left[\widehat{I}\right] = \exp\{-\Delta\}$, where $K \sim \mathrm{Poisson}(\lambda)$ and $(\tau_\star,\theta_\star)_j \sim P$ (built from the conditional and marginal). In fact, we sample $\tau_{\star j} \sim \mathrm{Uniform}[\eta,\eta']$. `Adame:` Note here, we have to define this measure properly. It should be a product measure.

In that case, we find

$$\widehat{I} = \exp\{(\lambda - c)\} \prod_{j=1}^{K} \frac{(c - \widehat{\Delta}(\tau_{\star j}))}{\lambda} \tag{4.12}$$

The choice of λ and c is crucial for the efficiency of the estimator. An optimal choice would be to take $\lambda = \sup_{\theta \in \Theta} \widehat{\Delta} - \inf_{\theta \in \Theta} \widehat{\Delta}$ and $c = \sup_{\theta \in \Theta} \widehat{\Delta}$. Refer to section A.1 for a detailed proof of unbiasedness.

In order to make proper use of this method with $\widehat{\Delta}(\eta, \eta')$, one would use a prior with finite support. We then suggest employing a standard-normal truncated at the 99% percentiles.

4.1.5 Divide-and-Conquer

$$p(\theta, \eta \mid D, D_0) \propto \frac{L(D \mid \theta) L_1(D_0 \mid \theta) \cdots L_\eta(D_0 \mid \theta) \pi_A(\eta) \pi_0(\theta)}{c_0(\theta)} \tag{4.13}$$

4.2 Pseudo-marginal via Importance Sampling

4.3 Other Approaches

4.3.1 Harmonic Estimator

5 Experiments

5.1 Assessing Exact Methods

`Adame:` Add the new estimators. Is it methodology? Because it is still an example.

5.1.1 Building an estimator

The most simple approach is the harmonic estimator, since

$$c_0(\eta) = \mathbb{E}_{\theta\sim\pi_0}[L_0(\theta \mid D_0)^\eta].$$

Then, we define δ_a as

$$\delta_a(\eta, \theta_1, \ldots, \theta_K) = \frac{1}{K}\sum_{i=1}^{K} L_0(\theta_i \mid D_0)^\eta \tag{5.1}$$

And with this estimator we can then employ a Portkey Barker's (Vats et al., 2021) to sample from the power posterior.

This δ_a has a major two-sided feature, while it is easy to sample from the initial prior, its efficiency is low due to its large non-informativeness.

In order to improve the efficiency, we can produce an importance sampling estimator by concentrating the probability density around the $\hat{\theta}_{MLE}$, which gives us

$$c_0(\eta) = \int_{\theta\in\Theta} L_0(\theta \mid D_0)^\eta \pi_0(\theta)\,\mathrm{d}\theta = \mathbb{E}_{\theta\sim\pi_{MLE}}\left[L_0(\theta \mid D_0)^\eta \frac{\pi_0(\theta)}{\pi_{MLE}(\theta)}\right].$$

We can then define δ_b as:

$$\delta_b(\eta, \theta_1, \ldots, \theta_K) = \frac{1}{K}\sum_{i=1}^{K} L_0(\theta_i \mid D_0)^\eta \frac{\pi_0(\theta)}{\pi_{MLE}(\theta)}. \tag{5.2}$$

`Adame:` Different compatibility scenarios.

5.1.2 The choice of λ and c for the Gaussian likelihood with known Σ.

Considering the functional form of our estimator $\widehat{\Delta}(\eta, \eta') = (\eta' - \eta)\log L_0\left(D_0 \mid \widetilde{\theta}\right)$ and that L_0 is the likelihood of $D_0 = \{d_{01}, \ldots, d_{0N}\}$ where each $d_{0j} \sim \mathcal{N}_d(\theta, \Sigma)$ for $j = 1, \ldots, N$ with known Σ, we can find that

$$\arg\max_{\theta\in\Theta} \log L_0(D_0 \mid \theta) = \hat{\theta}_{\mathrm{MLE}} = \frac{1}{N}\sum_{j=1}^{N} d_{0j},$$

and, therefore, we have that

$$\max_{\theta\in\Theta} \log L_0(D_0 \mid \theta) = -\frac{Nd}{2}\log(2\pi) - \frac{N}{2}\log\det\Sigma - \frac{N}{2}\operatorname{tr}(\Sigma^{-1}S),$$

where $S = \frac{1}{N}\sum_{j=1}^{N}(x_j - \hat{\theta}_{\mathrm{MLE}})(x_j - \hat{\theta}_{\mathrm{MLE}})^\top$

5.1.3 The choice of λ and c for the exponential family

`Adame:` Produce analytical results for exponential family. I have it somewhere.

A canonical exponential family has the following density:

$$L(X \mid \theta) = b(X)\exp\Big\{\theta^\top t(X) - c(\theta)\Big\},\ \theta \in \Theta \subseteq \mathbb{R}^d, \tag{5.3}$$

where Θ is open and connected.

5.2 Using Exact Methods as Baseline

`Adame:` Here I'd like to use the best exact method to assess the quality of the approximate methods

5.3 Numeric Experiments

5.3.1 Convergence to the baseline

We evaluate the proposed estimators under a multivariate Gaussian model with known covariance ($p = 3$, $\Sigma = I_3$, $\Sigma_0 = 2I_3$), placing weakly informative Beta$(1, 1)$ priors on η and a $\mathcal{N}_3(0, \Sigma_0)$ prior on θ. Historical and current datasets each contain $N_0 = N = 50$ observations. Three discrepancy scenarios are constructed by shifting the historical data mean by $\delta \in \{0.1, 0.5, 1.0\}$ per coordinate, representing low, moderate, and high conflict between historical and current information.

Rather than running each method for a fixed number of iterations, we adopt a convergence-based protocol: each estimator is run in independent batches of four chains and 5 000 iterations until the two-sample Kolmogorov–Smirnov (KS) distance between its accumulated posterior draws for η and the Stan HMC baseline drops below 0.05. The number of chains and total wall-clock time required to reach this threshold serve as our primary efficiency metrics. The exact algorithm—which uses the closed-form normalising

constant ratio—acts as a reference. We compare it against three approximate pseudo-marginal methods: the prior-importance estimator (δ_a), the MLE importance-sampling estimator (δ_b), and the power-posterior estimator.

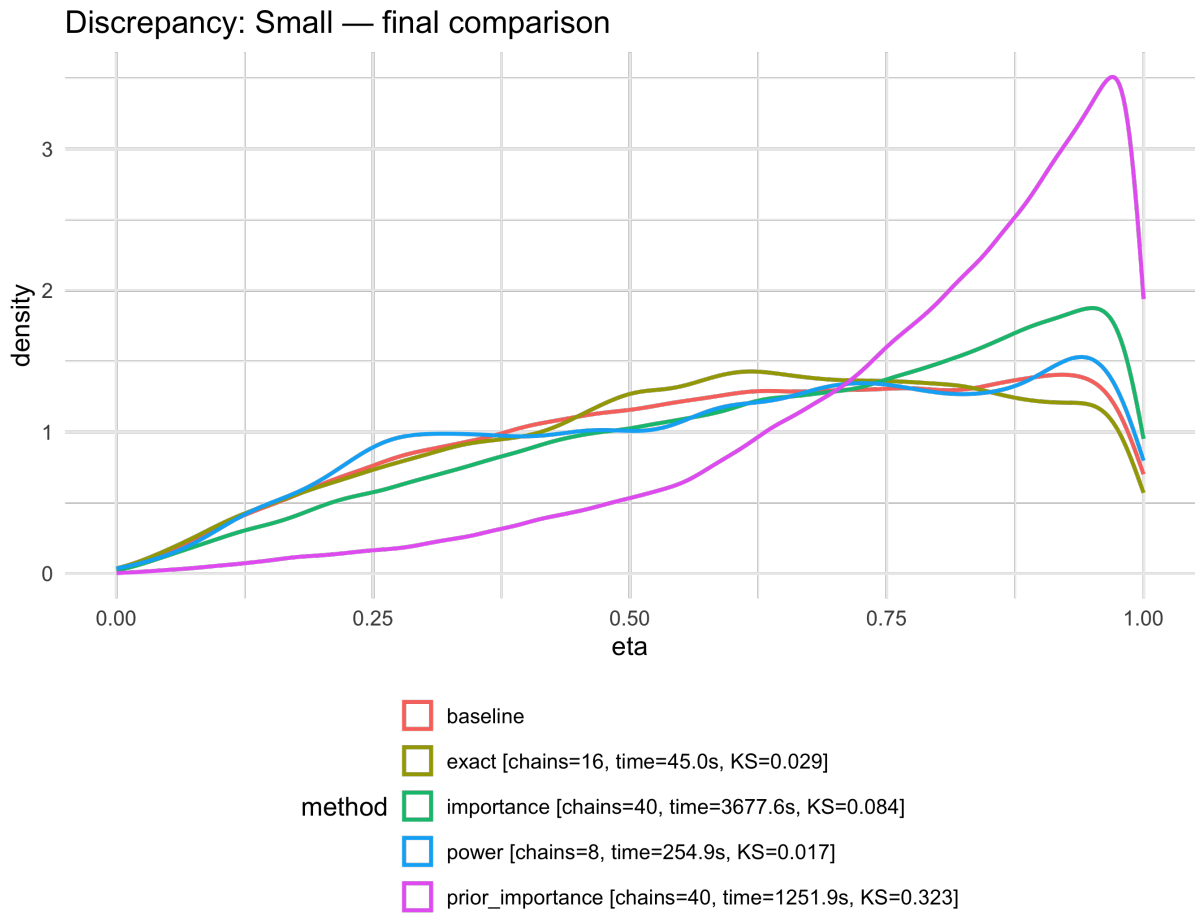

Figure 1 – Posterior densities for η under low discrepancy ($\delta = 0.1$). Each approximate estimator label reports the total number of chains run, wall-clock time, and final KS distance to the Stan baseline.

	Small ($\delta = 0.1$)		Medium ($\delta = 0.5$)		Large ($\delta = 1.0$)	
Method	Chains	KS	Chains	KS	Chains	KS
Exact	16	0.029	8	0.033	4	0.010
Prior-importance	40	0.323	40	0.229	4	0.027
Importance	40	0.084	12	0.050	4	0.026
Power	8	0.017	4	0.033	4	0.021

Table 1 – Total chains required and final KS distance to the Stan HMC baseline. A dash indicates the method reached 10 batches (40 chains) without satisfying the convergence criterion.

5.3.2 Discussion of results

The results reveal a consistent pattern across discrepancy regimes. Under low discrepancy (Figure 1), the posterior for η spreads across a wide range, reflecting genuine

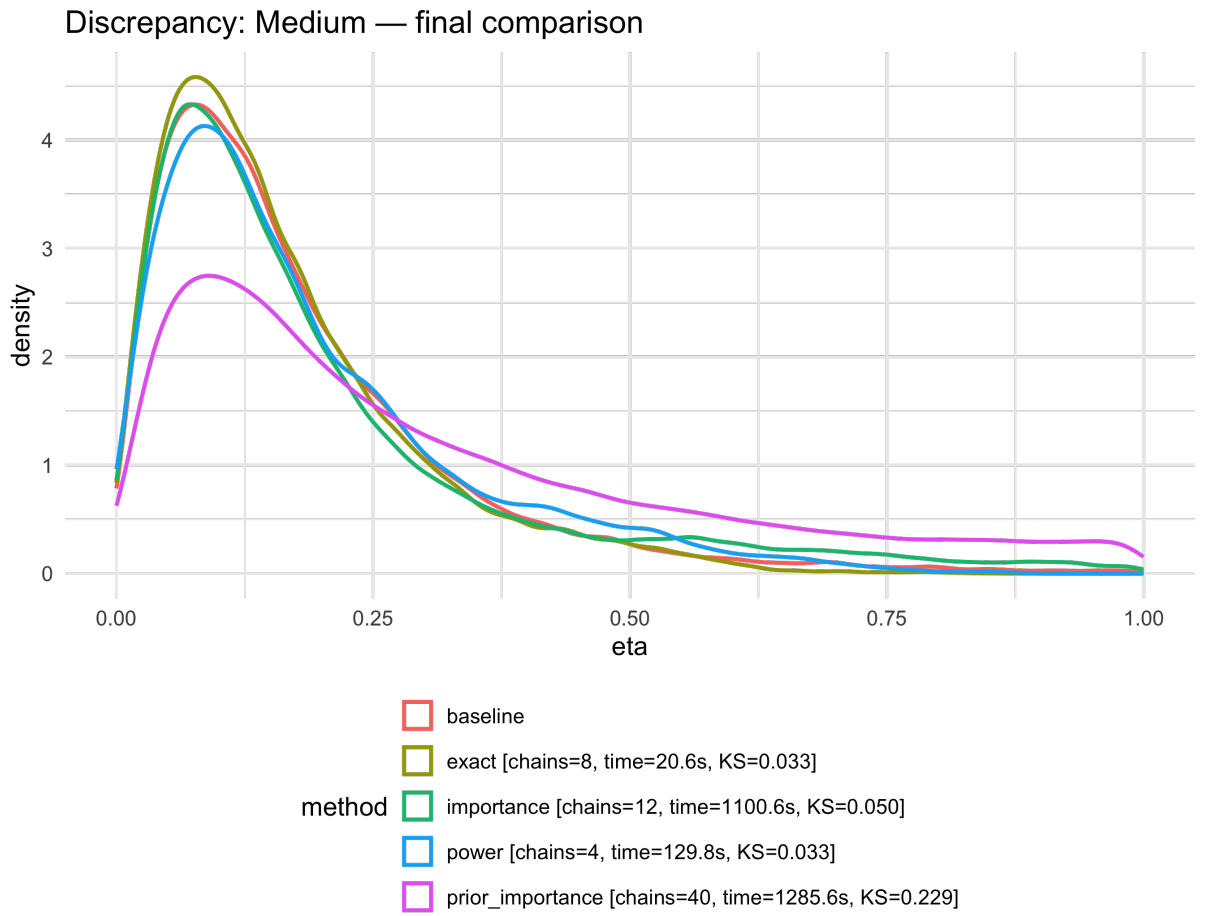

Figure 2 – Posterior densities for η under moderate discrepancy ($\delta = 0.5$).

uncertainty about how much weight to assign to the historical data. In this regime the estimators differ most sharply: the power estimator converges in just two batches (8 chains, KS= 0.017), the exact algorithm requires four batches (16 chains, KS= 0.029), whereas the importance estimator exhausts all ten batches without reaching the threshold (KS= 0.084) and the prior-importance estimator deviates substantially from the baseline (KS= 0.323). The inefficiency of δ_a here is expected: when the posterior for η is diffuse, prior samples for θ are a poor proposal for the normalising-constant integral, yielding high-variance estimates of the log-ratio.

Under moderate discrepancy (Figure 2), the posterior shifts toward smaller values of η, indicating the model has begun to down-weight the historical data. The power estimator again converges in a single batch (4 chains), the exact algorithm in two, and the importance estimator in three. The prior-importance estimator still fails to reach the threshold, though its KS distance has improved relative to the low-discrepancy case.

When the two datasets are in strong conflict (Figure 3), the posterior concentrates sharply near $\eta \approx 0$. In this regime all four estimators converge in a single batch of four chains, and the KS distances are uniformly small. The sharp concentration of the posterior makes the normalising-constant ratio easy to estimate regardless of the proposal, erasing

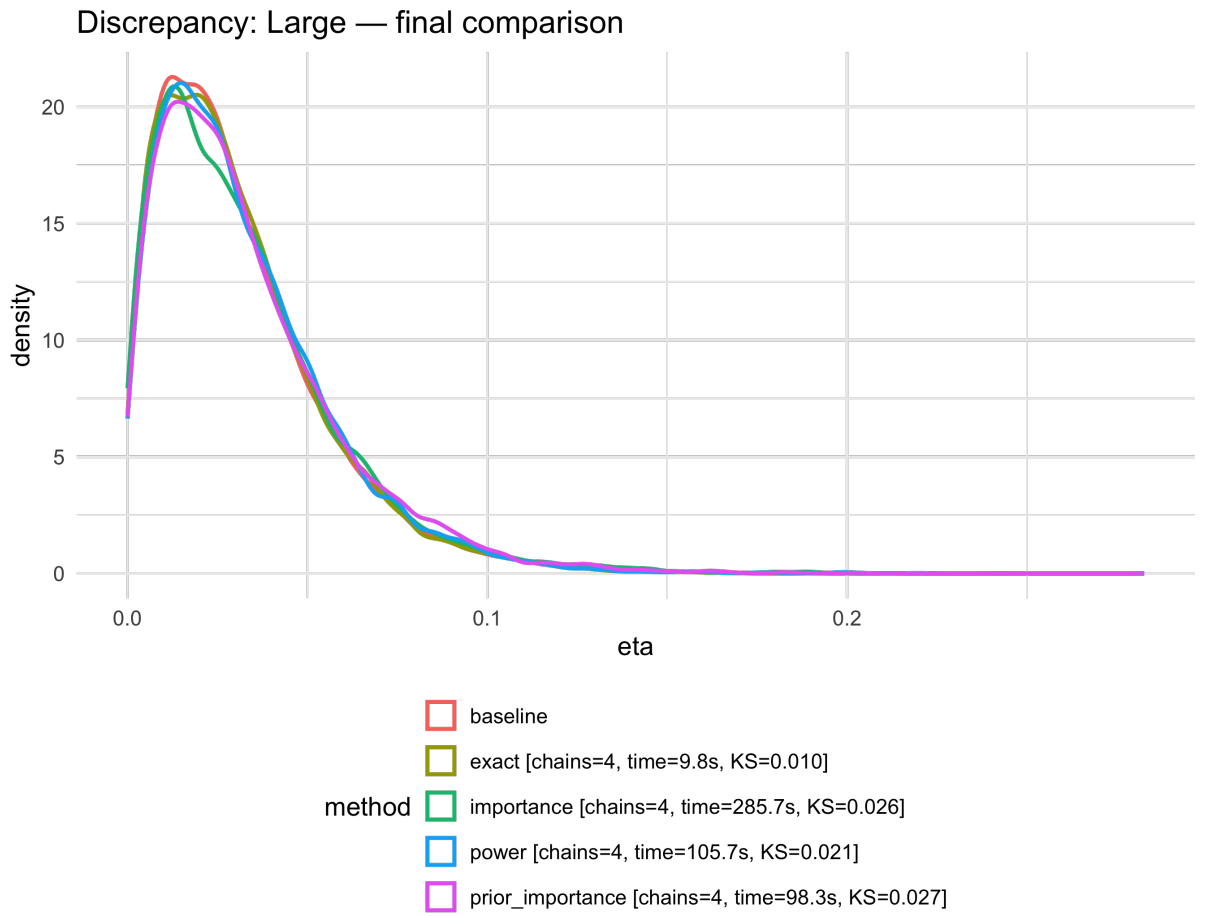

Figure 3 – Posterior densities for η under high discrepancy ($\delta = 1.0$). Under strong conflict between datasets the posterior concentrates near zero and all methods converge in a single batch.

the differences observed at lower discrepancy levels.

Overall, the power estimator is the most efficient of the three approximate methods: it matches or beats the exact algorithm in chains required across all scenarios. The prior-importance estimator is consistently the weakest, particularly when the posterior is diffuse. These findings suggest that the choice of estimator matters most in low-discrepancy settings, where the posterior for η is broad and the normalising constant must be estimated accurately over a wide range of values.

6 Discussion

Code Availability The code is available at:

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APPENDIX A – Proofs

A.1 Proof that 4.12 is unbiased

Proof. The result is immediate, we first take the expected value with respect to P that leaves us with the actual Δ inside the expected value.

$$\mathbb{E}_{K,(\tau_\star,\theta_\star)_1,\ldots,(\tau_\star,\theta_\star)_K}\left[\widehat{I}\right] = \mathbb{E}_{K,(\tau_\star,\theta_\star)_1,\ldots,(\tau_\star,\theta_\star)_K}\left[\exp\{(\lambda - c)\}\prod_{j=1}^{K}\frac{(c-\widehat{\Delta}(\tau_{\star j}))}{\lambda}\right], \tag{A.1}$$

$$= \mathbb{E}_K\left[\exp\{(\lambda - c)\}\prod_{j=1}^{K}\frac{(c-\Delta)}{\lambda}\right], \tag{A.2}$$

$$= \mathbb{E}_K\left[\exp\{(\lambda - c)\}\left(\frac{(c-\Delta)}{\lambda}\right)^K\right], \tag{A.3}$$

then, we take the expected value with respect to $K \sim \text{Poisson}(\lambda)$

$$\begin{aligned}
\mathbb{E}_{K,(\tau_\star,\theta_\star)_1,\ldots,(\tau_\star,\theta_\star)_K}\left[\widehat{I}\right] &= \sum_{i=0}^{\infty}\mathbb{P}[K=i]\exp\{\lambda - c\}\left(\frac{(c-\Delta)}{\lambda}\right)^i, \\
&= \sum_{i=0}^{\infty}\frac{e^{-\lambda}\lambda^i}{i!}\exp\{\lambda - c\}\left(\frac{(c-\Delta)}{\lambda}\right)^i, \\
&= e^{-c}\sum_{i=0}^{\infty}\frac{(c-\Delta)}{i!} = e^{-c}e^{c-\Delta} = e^{-\Delta}.
\end{aligned}$$

□